

## Numerical Methods

BCA — Semester IV — 2021 — Answer Sheet  
Subject Code: CACS252

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Group C Attempt any TWO questions [2x10=20]

2. Compute the root of equation  $(x^2 - 5x + 6 = 0)$  in the vicinity of  $(x = 5)$  using Newton-Raphson method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Estimate the value of  $(\ln(2.5))$  using Newton-Gregory forward difference formula given the following data: X1.02.03.04.0 Ln(x) 0.00.69311.09861.3863

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Write an algorithm to compute integration using Simpson's 1/3 rule. Evaluate integration of  $(\int_0^3 (3x^2 + 2x - 5) dx)$  using Simpson's 1/3 rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is meant by ill-conditioned system? Find the Cholesky decomposition of the following matrix:  $\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Use the classical RK method to estimate  $(y(0.4))$  of the equation  $(\frac{dy}{dx} = x^2 + y^2)$  with  $(y(0) = 0)$ , assume  $(h = 0.2)$ .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Solve the Poisson equation  $(\nabla^2 f = 2x^2 y^2)$  over the square domain  $(0 \leq x \leq 3 \text{ and } 0 \leq y \leq 3)$  with  $(f = 0)$  on the boundary and  $(h = 1)$ .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write a short note on (Any Two): a) Types of Errors b) Convergent Method c) PDE

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

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**9. Write an algorithm and program to compute root of nonlinear equation using bisection method.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**10. Given the data points:  $(0, 1), (2, 5), (4, 16), (6, 37), (8, 64)$  Estimate the function value of  $f(x)$  at  $x = 14$  using cubic splines.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**11. a) Solve the following system of equations using Gauss-Seidel method:  $2x - 7y - 10z = -17$ ,  $5x + y + 3z = 14$ ,  $x + 10y + 9z = 7$  b) Write an expansion of Taylor's Theorem. Solve the equation:  $y'(x) = x^2 + y^2$ ,  $y(0) = 1$  for  $x = 0.5$  using Taylor method.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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*These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.*